Why Have House Prices in Turkey Gone Up in the Last Ten Years?

Project Final Report
EMU660 Statistical Learning and Analytics

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Abstract

Housing affordability has emerged as one of the most pressing socioeconomic challenges in Türkiye over the past decade. This study investigates the key drivers of the sharp rise in housing prices between 2014 and 2024, drawing on four official monthly time-series datasets from the Central Bank of the Republic of Türkiye (TCMB/EVDS): the House Price Index (HPI), housing sales, USD/EUR exchange rates, and credit card sectoral expenditure. Through exploratory analysis, correlation testing, regional comparison, and log-log regression, we find that exchange rate depreciation and broad inflation are the dominant macroeconomic forces behind rising house prices, while sales volumes display an inverse relationship with price levels. Regional heterogeneity is substantial, and our model explains approximately 95–99% of the variance in the national HPI.

1 Introduction

Housing has become one of the most consequential economic and social issues in Türkiye over the past ten years. The national House Price Index (HPI) accelerated dramatically from around 2019–2020, placing homeownership out of reach for large segments of the population. This report addresses the question:

Why have house prices in Turkey gone up in the last ten years?

We combine four monthly time-series from the EVDS of the Central Bank of Türkiye (TCMB): the HPI, housing sales, USD/TRY and EUR/TRY exchange rates, and credit card sectoral expenditure. Section 2 describes the data, Section 3 presents the analysis, and Section 4 summarises the findings.

2 Data

2.1 Data Source

All data were obtained from the **Electronic Data Delivery System (EVDS)** of the Central Bank of the Republic of Türkiye (TCMB). The project draws on four datasets:

- **House Price Index (HPI)** — monthly index at national and regional (NUTS2) levels.
- **House Sales Statistics** — monthly provincial sales data.
- **Exchange Rates (USD/TRY, EUR/TRY)** — monthly series.
- **Credit Card Sectoral Expenditures** — monthly sector-based spending.

2.2 General Information About the Data

The processed dataset consists of three main analytical tables described below.

2.2.1 Exchange Rate Table

Monthly USD/TRY and EUR/TRY rates used as indicators of macroeconomic conditions and cost-side pressures. Both series trend sharply upward, with the most dramatic depreciation occurring after 2021.

2.2.2 Credit Card Expenditure Table

Monthly credit card spending by sector (millions of TRY), used as a broad nominal consumption proxy. As shown in Table 2, the largest categories are e-commerce, supermarkets, food, clothing, and fuel.

2.2.3 Housing Panel Dataset

The main analytical table: Year, Month, Region (NUTS2), House Price Index, and Total Sales. The national Türkiye aggregate is the primary object of regression.

2.3 Preprocessing

Raw EVDS files (Excel, wide format, province-level) were preprocessed in Python using `pandas`: dates standardised, data reshaped to long format, coded column names (e.g., `KT2`, `TR62`) translated, and province sales aggregated to NUTS2.

Note: portions of the preprocessing code were developed with AI assistance. All R analysis, interpretation, and conclusions are the authors' own work.

3 Analysis

3.1 Exploratory Data Analysis

Table 1 presents descriptive statistics for each key variable.

Table 1: Summary statistics of key variables.

Variable	Min	Mean	Median	Max	SD
House Price Index	7.7	51.1	14.6	204.4	60.8
Monthly Sales (units)	45,218.0	126,140.7	120,437.5	265,223.0	35,718.6
USD/TRY Rate	2.1	12.6	6.1	42.7	12.4
EUR/TRY Rate	2.7	13.9	6.8	49.9	13.7

The HPI rose from ~8 to over 200 by end of sample. Exchange rates followed a similar upward trajectory; sales were far more volatile, suggesting transaction volume decoupled from price levels.

Table 2: Top 8 credit card expenditure categories — cumulative total (2014–2025).

Expense Category	Total Spending (Billion TRY)
E-Alışveriş	15,913,640
Market ve Alışveriş Merkezleri	10,409,810
Çeşitli Gıda	3,892,815
Giyim ve Aksesuar	3,889,505
Benzin ve Yakıt İstasyonları	3,542,543
Hizmet Sektörleri	3,486,972
Elektrik-Elektronik Eşya / Bilgisayar	3,308,526
Yemek	3,156,894

Credit card spending is used as a broad nominal inflation proxy rather than a direct measure of housing demand.

3.2 Trend Analysis

3.2.1 House Price Index Over Time

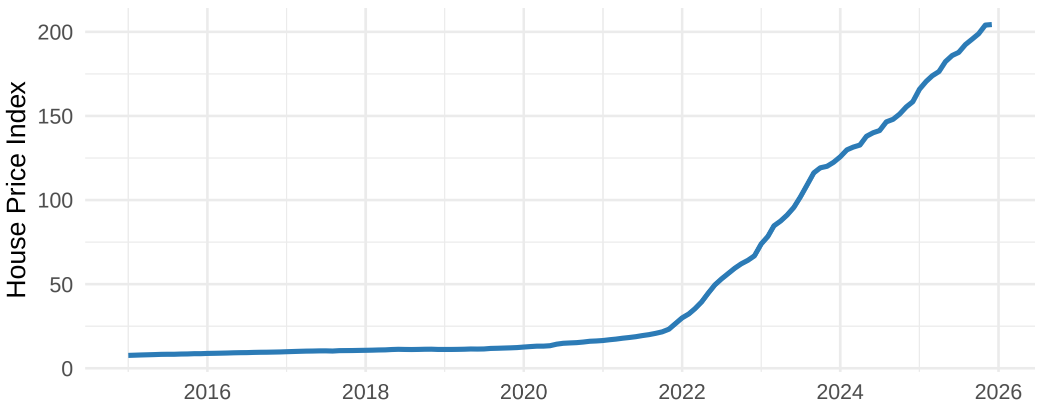

Figure 1: National House Price Index over time (2014–2025).

The HPI was stable until 2019, then accelerated sharply through 2021–2023 (Figure 1), coinciding with severe exchange rate depreciation and high inflation — consistent with a cost-push interpretation.

3.2.2 Year-over-Year Growth Rate Comparison

Year-over-year rates are a more robust diagnostic than raw levels: they reveal whether currency shocks and price jumps coincide.

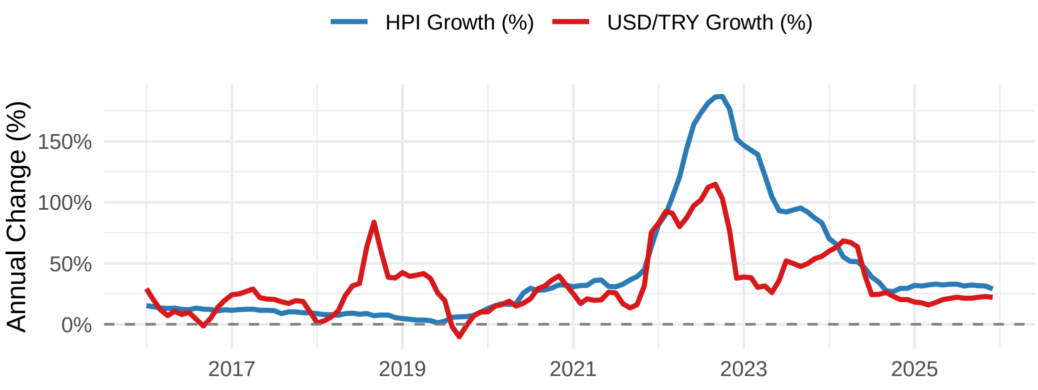

Figure 2: Year-over-year growth rates: House Price Index vs. USD/TRY.

Figure 2 shows that USD/TRY and HPI growth spikes align closely, especially post-2020, supporting a connection between currency shocks and house-price inflation — though this reflects co-movement, not a causal estimate.

3.2.3 Exchange Rates Over Time

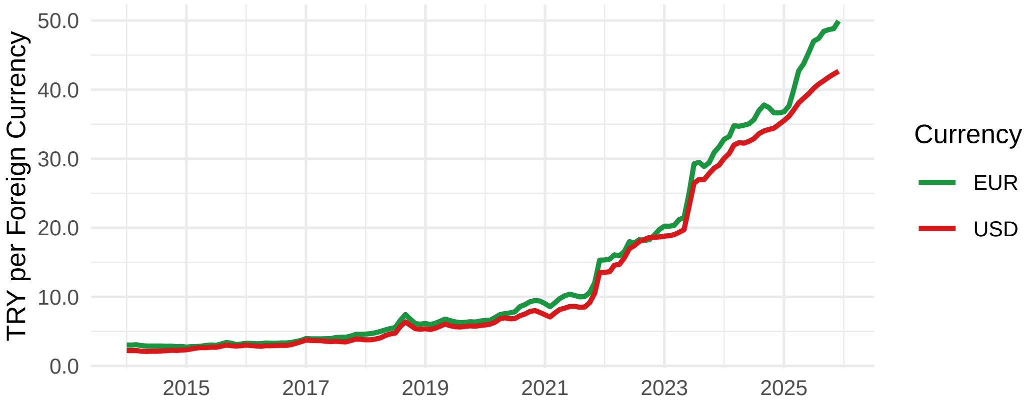

Figure 3: USD/TRY and EUR/TRY exchange rates over the sample period.

3.2.4 Annual Housing Sales

Sales fluctuated without a sustained upward trend (Figure 4), a first indication that price growth was not primarily demand-driven.

3.3 Correlation Analysis

3.3.1 Correlation Heatmap

Figure 5 summarises pairwise correlations between all key variables.

Table 3: Pearson correlations with the House Price Index.

Variable Pair	Pearson r
HPI vs USD/TRY	0.990
HPI vs EUR/TRY	0.990

HPI vs CC Spend	0.973
HPI vs Sales	0.184

The HPI correlates strongly with exchange rates ($r \approx 0.99$) and credit card spending ($r \approx 0.97$), but weakly with sales ($r \approx 0.18$, Table 3). Shared non-stationarity may inflate levels-based correlations; the YoY comparison and regression models address this.

3.3.2 Scatter Plot: log(HPI) vs. log(USD/TRY)

Figure 6 confirms a tight log-linear relationship at all exchange rate levels, not merely a shared-trend artefact.

3.4 Regional Comparison

3.4.1 Regional HPI Trajectories

3.4.2 Regional Growth Ranking

Coastal and western regions lead in percentage growth (Table 4), driven by migration, tourism, and second-home demand. İstanbul leads in absolute index level. All regions exceed 2,000% nominal appreciation — the surge is nationwide.

3.5 Model Fitting

We fit log-log regression models, where coefficients approximate elasticities. Given the strong upward trends in all series, results are interpreted as associational rather than causal.

3.5.1 Baseline Model — Exchange Rate Only

$$\log(\text{HPI}_t) = \beta_0 + \beta_1 \log(\text{USD/TRY}_t) + \varepsilon_t$$

Table 5: Baseline model coefficients. $R^2 = 0.959$.

	Term	Estimate	Std. Error	t value	p value
(Intercept)	Intercept	0.644	0.052	12.49	0.0000
log_USD	log(USD/TRY)	1.200	0.022	55.03	0.0000

USD/TRY alone explains ~96% of log(HPI) variation. The elasticity $\hat{\beta}_1 \approx 1.20$ implies a 1% exchange rate increase is associated with a 1.20% rise in the HPI.

3.5.2 Full Model — Exchange Rate, Sales, and Credit Card Expenditure

$$\log(\text{HPI}_t) = \beta_0 + \beta_1 \log(\text{USD}_t) + \beta_2 \log(\text{Sales}_t) + \beta_3 \log(\text{CC}_t) + \varepsilon_t$$

Table 6: Full model coefficients. $R^2 = 0.986$, Adjusted $R^2 = 0.985$.

	Term	Estimate	Std. Error	t value	p value	Sig
(Intercept)	Intercept	-8.956	0.820	-10.92	0.0000	***
log_USD	log(USD/TRY)	0.256	0.062	4.12	0.0001	***
log_Sales	log(Sales)	-0.120	0.045	-2.68	0.0084	**
log_CC	log(CC Spend)	0.679	0.044	15.55	0.0000	***

The full model reaches adjusted $R^2 \approx 0.985$. USD/TRY remains dominant; the negative sales coefficient reflects demand compression at high prices; and the CC coefficient proxies broad nominal inflation.

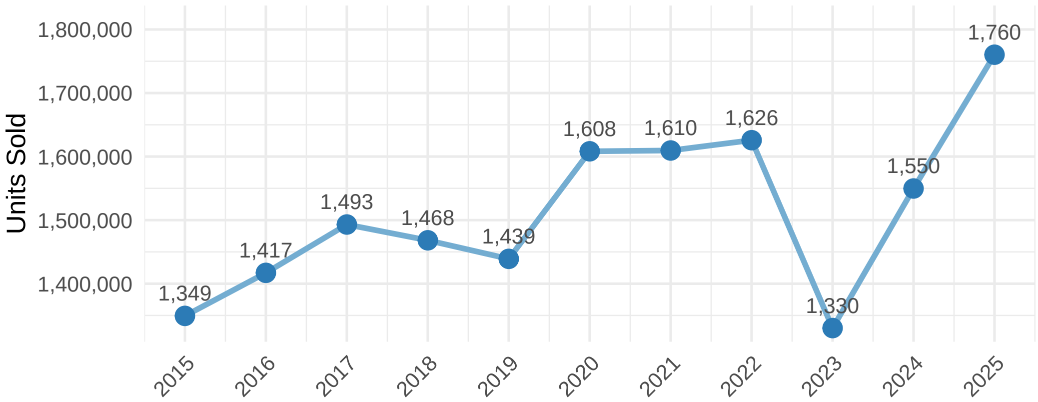

Figure 4: Annual housing sales for Türkiye. Labels show thousands of units sold.

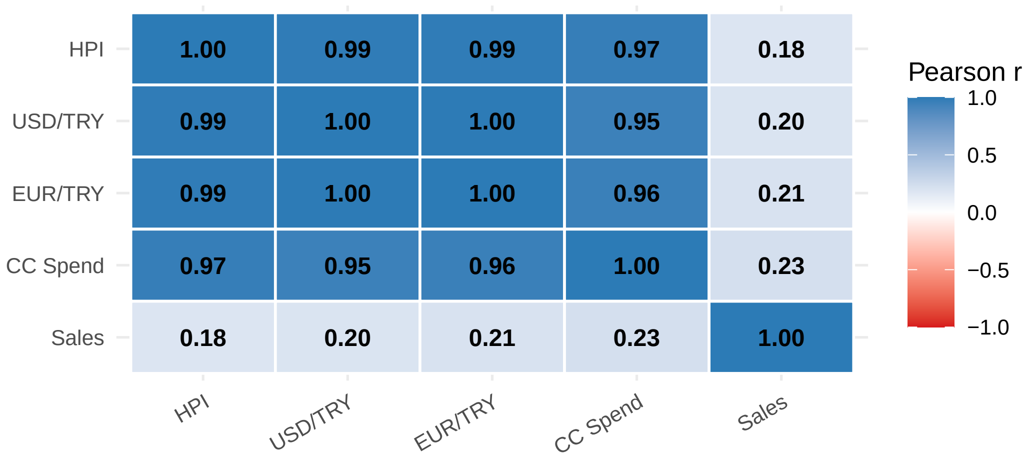

Figure 5: Pearson correlation heatmap of key variables.

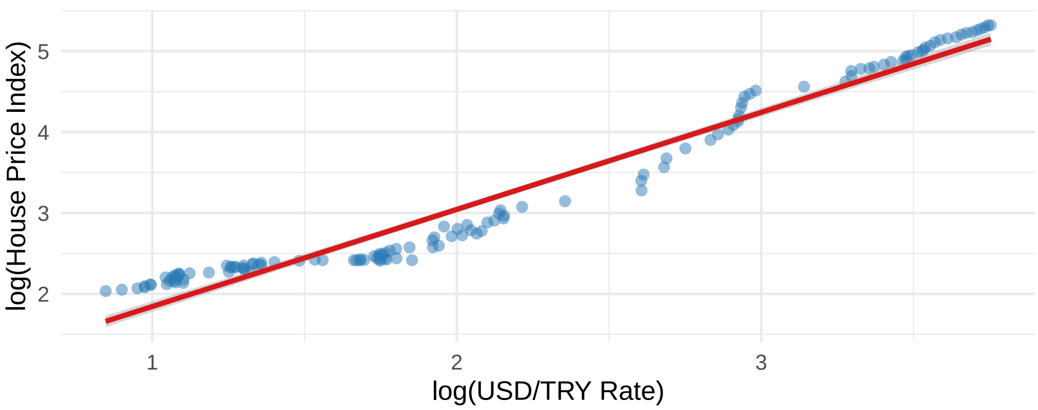

Figure 6: Scatter plot of log(HPI) against log(USD/TRY). Each point is one monthly observation.

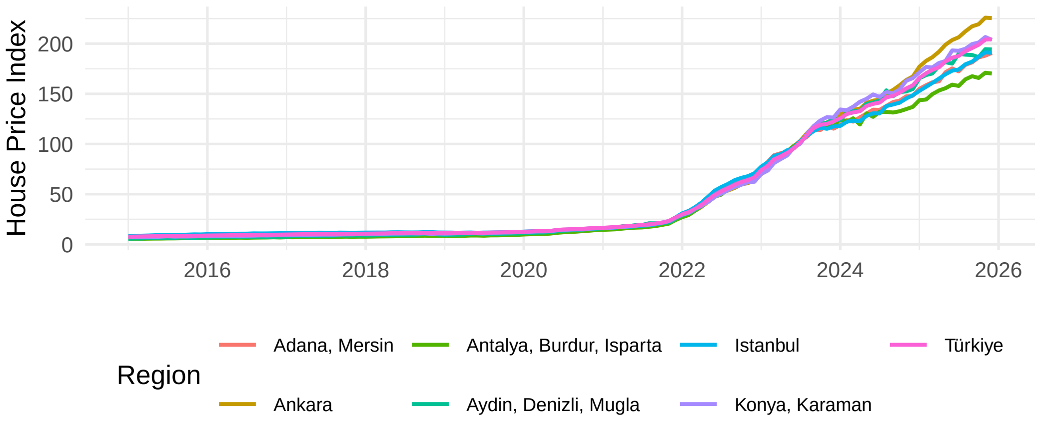

Figure 7: House Price Index trajectories for selected regions (2014–2025).

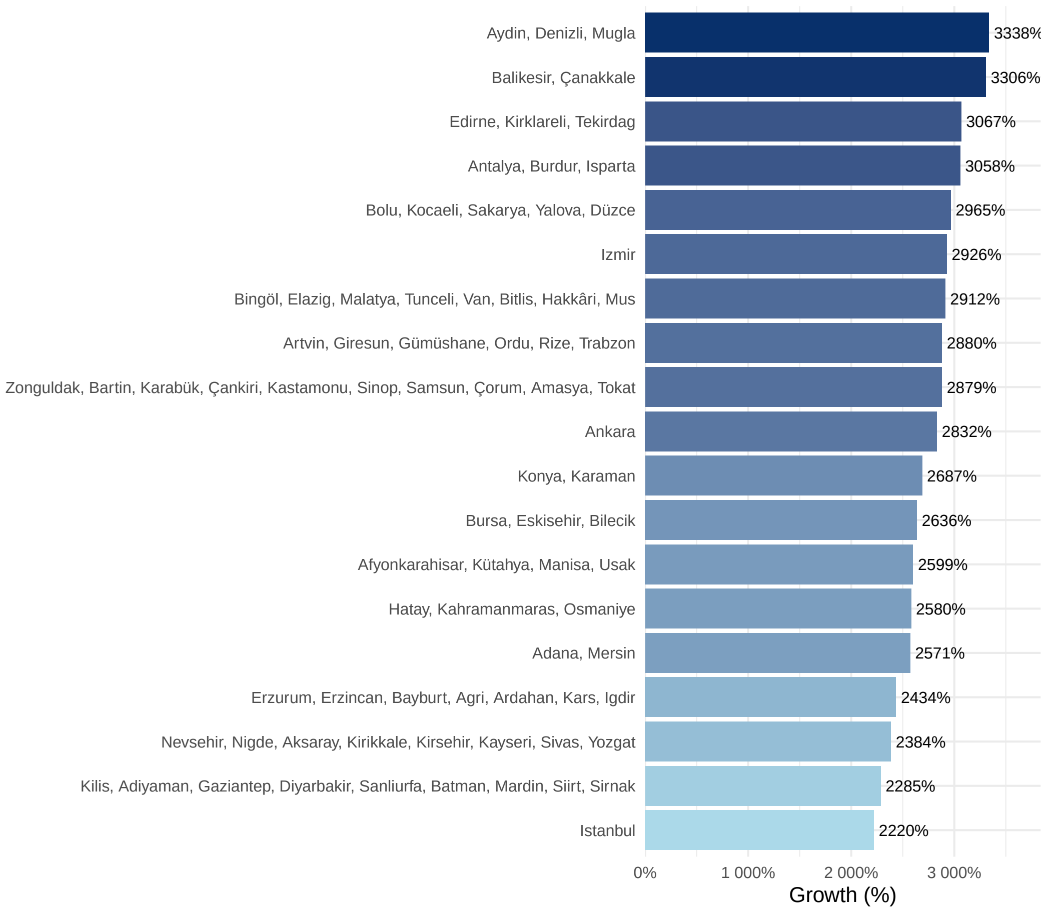

Figure 8: Total HPI percentage growth by NUTS2 region (first to latest observation).

Table 4: HPI growth by NUTS2 region (first to latest observation).

Region	First HPI	Latest HPI	Growth (%)
Aydın, Denizli, Muğla	5.7	194.2	3337.5
Balıkesir, Çanakkale	6.2	211.9	3306.1
Edirne, Kırklareli, Tekirdağ	6.2	196.1	3067.2
Antalya, Burdur, Isparta	5.4	170.2	3057.5
Bolu, Kocaeli, Sakarya, Yalova, Düzce	6.9	210.9	2964.8
İzmir	6.6	198.8	2926.5
Bingöl, Elazığ, Malatya, Tunceli, Van, Bitlis, Hakkâri, Muş	8.2	245.8	2912.4
Artvin, Giresun, Gümüşhane, Ordu, Rize, Trabzon	7.4	219.7	2880.3
Zonguldak, Bartın, Karabük, Çankırı, Kastamonu, Sinop, Samsun, Çorum, Amasya, Tokat	7.3	217.8	2879.1
Ankara	7.7	225.5	2831.9
Konya, Karaman	7.3	203.8	2687.3
Bursa, Eskişehir, Bilecik	7.4	203.0	2636.4
Afyonkarahisar, Kütahya, Manisa, Uşak	8.5	228.9	2599.2
Hatay, Kahramanmaraş, Osmaniye	8.1	216.3	2580.2
Adana, Mersin	7.1	190.7	2571.4
Türkiye	7.7	204.4	2571.4
Erzurum, Erzincan, Bayburt, Ağrı, Ardahan, Kars, Iğdır	9.8	247.8	2433.7
Nevşehir, Niğde, Aksaray, Kırıkkale, Kırşehir, Kayseri, Sivas, Yozgat	8.8	219.1	2384.5
Kilis, Adıyaman, Gaziantep, Diyarbakır, Şanlıurfa, Batman, Mardin, Siirt, Şırnak	8.8	210.4	2285.4
İstanbul	8.2	190.9	2219.8

3.5.3 Actual vs. Fitted Values

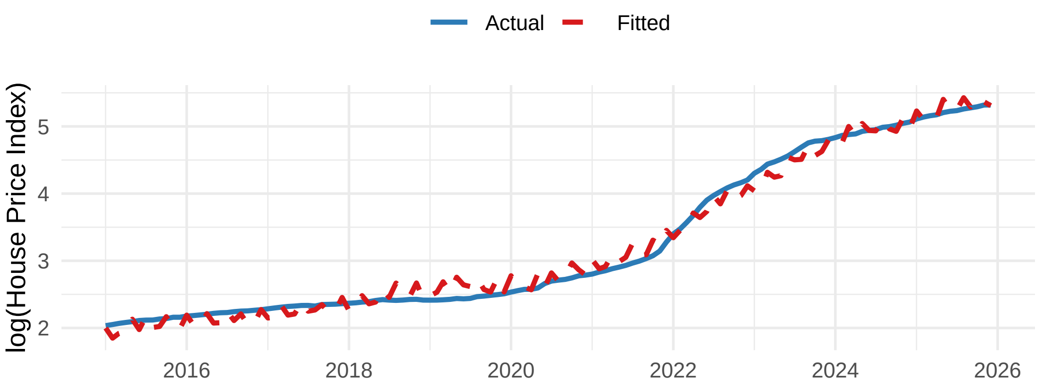

Figure 9: Actual and fitted log(HPI) — full model.

Fitted values closely track the actual series (Figure 9). Larger residuals in 2022 suggest mild non-linearity during the most extreme TRY depreciation.

3.5.4 Residual Plot

Residuals are small and near zero across most of the range (Figure 10), with slight curvature at the upper end corresponding to 2022–2023.

4 Results and Key Takeaways

1. **House prices surged after 2020**, with the HPI more than doubling during 2021–2023.
2. **Exchange rate depreciation is the strongest correlate** ($r \approx 0.99$), with a baseline elasticity of ~1.20.
3. **Sales volume does not explain the increase** — volatile and trending separately from prices, consistent with affordability-driven demand compression.
4. **Credit card expenditure proxies broad inflation**, not direct housing demand.

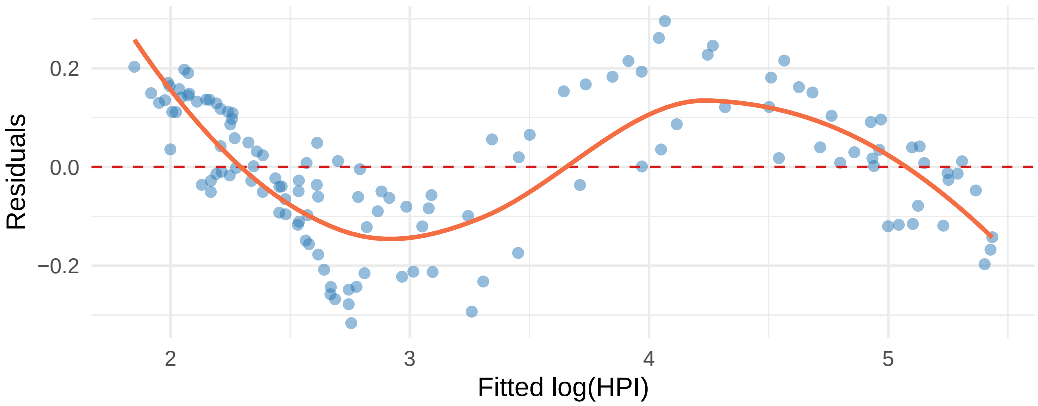

Figure 10: Residuals vs. fitted values — full model. The LOESS smoother (orange) reveals mild non-linearity at high fitted values.

5. **The surge is nationwide but regionally uneven**: coastal/western regions lead in percentage growth; İstanbul leads in absolute index level.
6. **Interpretation should remain cautious**: non-stationary variables share common trends, so results reflect association, not proven causality.
7. **Future work** incorporating real prices, interest rates, mortgage volumes, and time-series cointegration methods would sharpen causal inference.

5 Conclusion

Exchange rate depreciation and broad nominal inflation emerge as the central drivers of the Turkish housing price surge. USD/TRY alone explains ~96% of the variation in log(HPI); the full model reaches ~99%. These findings suggest that housing-market interventions in isolation — zoning changes, sales caps — are unlikely to be sufficient without broader macroeconomic stabilisation. The main limitation is that all variables are nominal and non-stationary; future work with real prices, interest rates, and cointegration methods would strengthen causal claims.

References

Central Bank of the Republic of Türkiye (TCMB). (2024). *Electronic Data Delivery System (EVDS)*. Retrieved from https://evds2.tcmb.gov.tr/